RPG and LLM

NEUGC
Northeast User Groups Conference

Jack Woehr
Seiden Group

SEIDEN GROUP

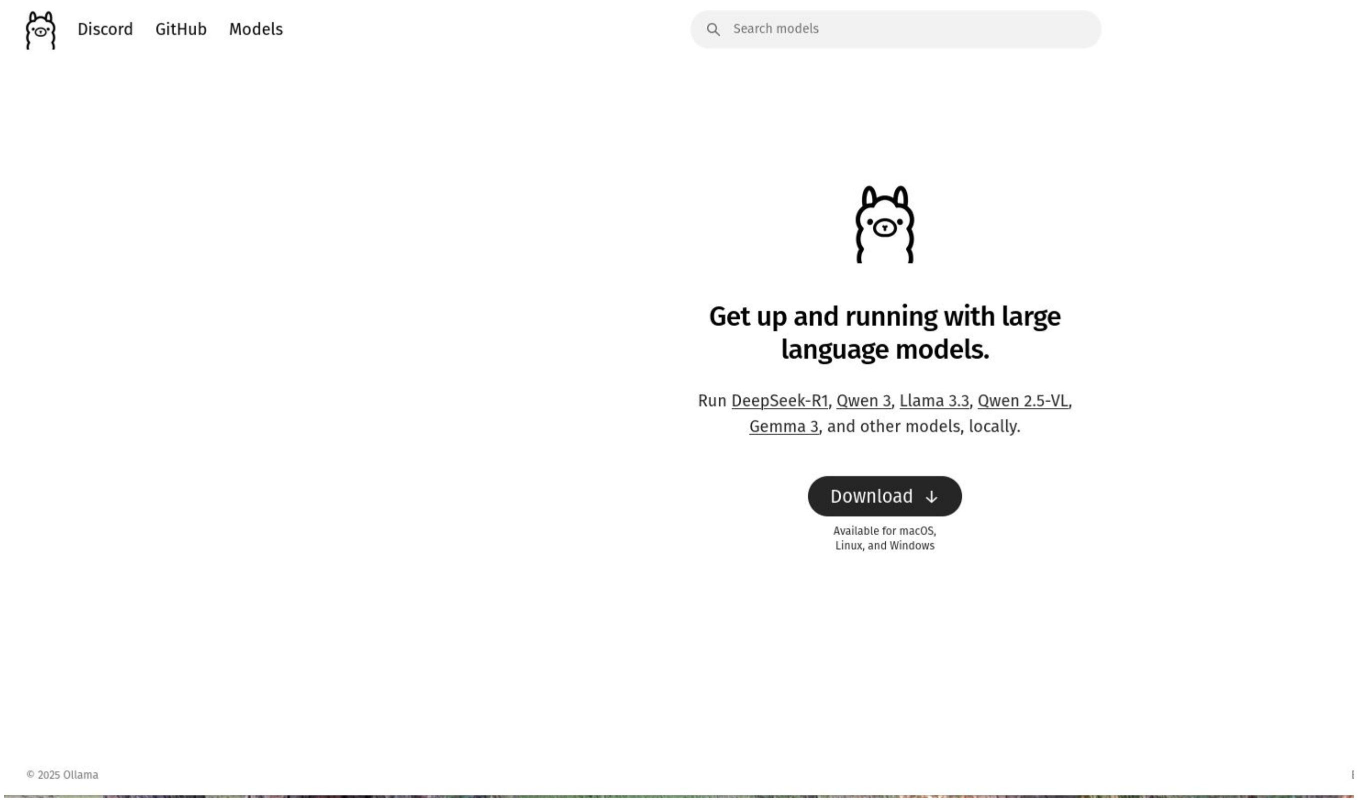
RPG

pre-columbian ceramic llama
Walters Art Museum

Our Agenda

1. Running models
2. Calling from `curl`
3. Calling from SQL
4. Calling from Python
5. IBM OSS DbToo
6. Early easy SQLRPGLE
7. SQLRPGLE with DSPF

[Ollama]


Discord
GitHub
Models
Search models
Get up and running with large language models.
Run DeepSeek-R1, Qwen 3, Llama 3.3, Qwen 2.5-VL, Gemma 3, and other models, locally.
Download
Available for macOS, Linux, and Windows
© 2025 Ollama

DMI: Dell Inc. Precision Tower 5810/0K240Y

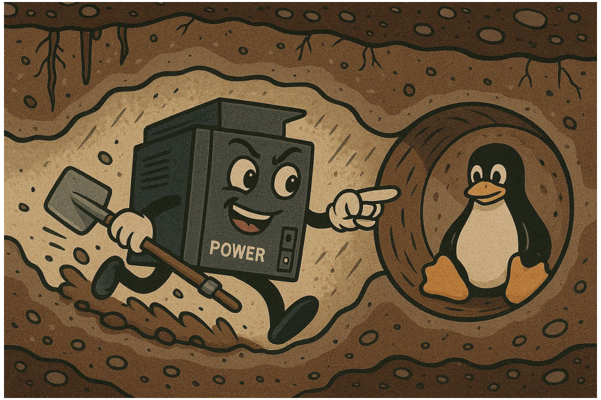
```
$ lscpu -p
# The following is the parsable format, which can be fed to other
# programs. Each different item in every column has an unique ID
# starting usually from zero.
# CPU,Core,Socket,Node,,L1d,L1i,L2,L3
0,0,0,0,,0,0,0,0
1,1,0,0,,1,1,1,0
2,2,0,0,,2,2,2,0
3,3,0,0,,3,3,3,0
4,4,0,0,,4,4,4,0
5,5,0,0,,5,5,5,0
6,6,0,0,,6,6,6,0
7,7,0,0,,8,8,8,0
8,8,0,0,,9,9,9,0
9,9,0,0,,10,10,10,0
10,10,0,0,,11,11,11,0
11,11,0,0,,12,12,12,0
12,12,0,0,,13,13,13,0
13,13,0,0,,14,14,14,0
14,0,0,0,,0,0,0,0
15,1,0,0,,1,1,1,0
16,2,0,0,,2,2,2,0
17,3,0,0,,3,3,3,0
18,4,0,0,,4,4,4,0
19,5,0,0,,5,5,5,0
20,6,0,0,,6,6,6,0
21,7,0,0,,8,8,8,0
22,8,0,0,,9,9,9,0
23,9,0,0,,10,10,10,0
24,10,0,0,,11,11,11,0
25,11,0,0,,12,12,12,0
26,12,0,0,,13,13,13,0
27,13,0,0,,14,14,14,0
```

```
$ nvidia-smi
Sat Jul 19 10:54:17 2025
+---------------------------------------------------------------------------------------+
| NVIDIA-SMI 535.247.01             Driver Version: 535.247.01   CUDA Version: 12.2     |
|-----------------------------------------+----------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                      |               MIG M. |
|=========================================+======================+======================|
|   0  NVIDIA GeForce GTX 1650        Off | 00000000:02:00.0 Off |                  N/A |
| 27%   29C    P8               5W /  75W |     49MiB /  4096MiB |      0%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+

+---------------------------------------------------------------------------------------+
| Processes:                                                                            |
|  GPU   GI   CI        PID   Type   Process name                            GPU Memory |
|        ID   ID                                                             Usage      |
|=======================================================================================|
|    0   N/A  N/A      2132      G   /usr/lib/xorg/Xorg                           39MiB |
|    0   N/A  N/A      2318      G   /usr/bin/gnome-shell                          5MiB |
+---------------------------------------------------------------------------------------+

$ lsmem
RANGE                                  SIZE  STATE REMOVABLE BLOCK
0x0000000000000000-0x000000007fffffff   2G online       yes     0
0x0000000100000000-0x00000014ffffffff  80G online       yes  2-41

Memory block size:         2G
Total online memory:      82G
Total offline memory:      0B
```

Reverse tunnel to Linux from IBM i

- Allow ourselves to use Ollama running on the Linux server from IBM i making the port appear local.
  - Necessary because the Linux server does not have a publicly accessible IP address.
- Leave this up as long as we are going to use Ollama from IBM i.

```
ssh -R 11434:localhost:11434 -l myuserprof mysystem.mycompany.com
```



`curl` the query

```
curl http://localhost:11434/api/chat -d '{
  "model": "llama3.2:3b",
  "stream": false,
  "messages": [
   { "role": "user", "content": "why do dogs bark?"}
  ]
}'
```

Patrick Behr calls ChatGPT from SQL using HTTP_POST

```
CREATE OR REPLACE FUNCTION pbehr.askChatGpt
(
    in_prompt      varchar(1024)
  , in_sysmsg      varchar(512)    default null
  , in_temp        varchar(10)     default '1'
  , in_maxtokens   varchar(10)     default '500'
)
RETURNS VARCHAR(4096)

BEGIN

    DECLARE apiUrl    VARCHAR(64)   DEFAULT
'https://api.openai.com/v1/chat/completions';
    DECLARE apiKey    VARCHAR(64)   DEFAULT
'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';
    DECLARE model     VARCHAR(32)   DEFAULT 'gpt-3.5-turbo';
    DECLARE headers   VARCHAR(256)  DEFAULT '';
    DECLARE messages  VARCHAR(2500) DEFAULT '';
    DECLARE sysmsg    VARCHAR(1000) DEFAULT '';
```

```
    DECLARE userMsg   VARCHAR(1500) DEFAULT '';
    DECLARE request   VARCHAR(4096) DEFAULT '';

    SET headers = '{"headers":{"Authorization":
"Bearer ' CONCAT apiKey CONCAT '",

"Content-Type":"application/json"}}';

IF in_sysmsg IS NOT NULL THEN
        SET sysmsg = '{"role":"system",
"content":"' CONCAT in_sysmsg CONCAT '"},';
    END IF;

    SET userMsg = '{"role":"user", "content":"'
CONCAT in_prompt CONCAT '"}';

    SET messages = '[' CONCAT sysmsg CONCAT
userMsg CONCAT ']';
```

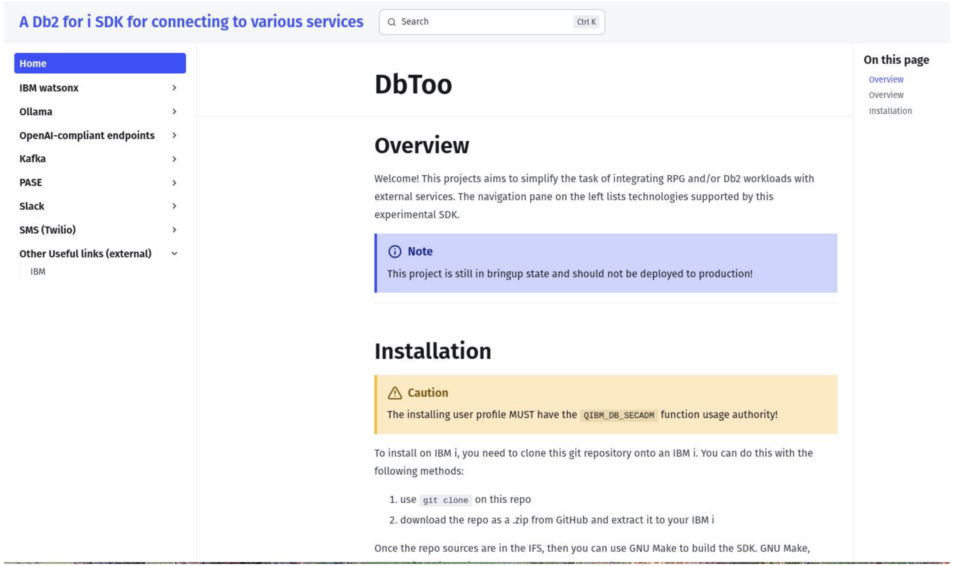
```
     SET request = '{"model":"'      CONCAT
model         CONCAT '",' CONCAT
                   '"temperature":' CONCAT
in_temp       CONCAT ','  CONCAT
                   '"max_tokens":'  CONCAT
in_maxtokens  CONCAT ','  CONCAT
                   '"messages":'    CONCAT
messages CONCAT '}';

    RETURN (
        SELECT apiResponse
        FROM JSON_TABLE(
            HTTP_POST( apiUrl, request, headers
),
            'lax $.choices[0]'
            COLUMNS(
                apiResponse varchar(4096) PATH
'lax $.message.content'
            )
        )
    );

END;
```

Same sort of thing in Python

```python
import ollama
# response = ollama.chat(model='llama3.1:8b', messages=[
# response = ollama.chat(model='granite-code:8b', messages=[
response = ollama.chat(model='llama3.2:3b', messages=[
  {
    'role': 'user',
    'content': 'Why is the sky blue?',
  },
])
print(response['message']['content'])
```

DbToo

- Code: https://github.com/IBM/AI-SDK-Db2-IBMi/
- Docs: https://ibm.github.io/AI-SDK-Db2-IBMi/



Do it in DbToo

```sql
set
   dbsdk_v1.ollama_protocol = 'http';
set
   dbsdk_v1.ollama_server = 'localhost';
set
   dbsdk_v1.ollama_port = 11434;
set
   dbsdk_v1.ollama_model = 'llama3.2:3b';
select
   dbsdk_v1.ollama_getmodel ()
from
   SYSIBM.SYSDUMMY1;
```

```sql
select
   dbsdk_v1.ollama_generate (
      prompt => '{ "role": "user", "content": "why do dogs
bark?"}') as myanswer
from
   SYSIBM.SYSDUMMY1;
set
   dbsdk_v1.ollama_model = 'granite-code:8b';
select
   dbsdk_v1.ollama_generate (
      prompt => '{ "role": "user", "content": "why do dogs
bark?"}') as myanswer
from
   SYSIBM.SYSDUMMY1;
```

Do it in SQLRPGLE (v-e-r-y simply)

```
**FREE
// Jack Woehr jack@seidengroup.com 2025-06-09
Ctl-Opt DftActGrp(*No) Option(*Srcstmt : *NodebugIO) main(qOllama);

dcl-s MyAnswer SQLTYPE(clob:16000000);
dcl-s DspStr varchar(52);

dcl-proc qOllama;

 exec sql call dbsdk_v1.ollama_setmodelforme(model => 'granite-code:8b');
 exec sql select dbsdk_v1.ollama_generate(prompt => '{ "role": "user", "content": "why do dogs bark?"}' ) into :MyAnswer
from SYSIBM.SYSDUMMY1;
 DspStr = MyAnswer_DATA;
 dsply DspStr;

 *inlr=*on;
end-Proc;
```

Roo, based on my example

```
**FREE
// Jack Woehr jack@seidengroup.com 2025-09-26
// New RPG program to interact with an Ollama-compatible LLM API
// QSYS/CRTSQLRPGI OBJ(NEWLLM) SRCSTMF('/home/jwoehr/work/AI/COMMONSweden25Q4/code/rpgollama/NEWLLM.SQLRPGLE') TEXT('Roo-authored take on RPGOLLAMA') CVTCCSID(*JOB)
Ctl-Opt DftActGrp(*No) Option(*Srcstmt : *NodebugIO) main(qNewLLM);

dcl-s MyAnswer SQLTYPE(clob:16000000);
dcl-s DspStr varchar(52);
dcl-s PromptText varchar(200) inz('{ "role": "user", "content": "Tell me a short story about a brave knight."}');

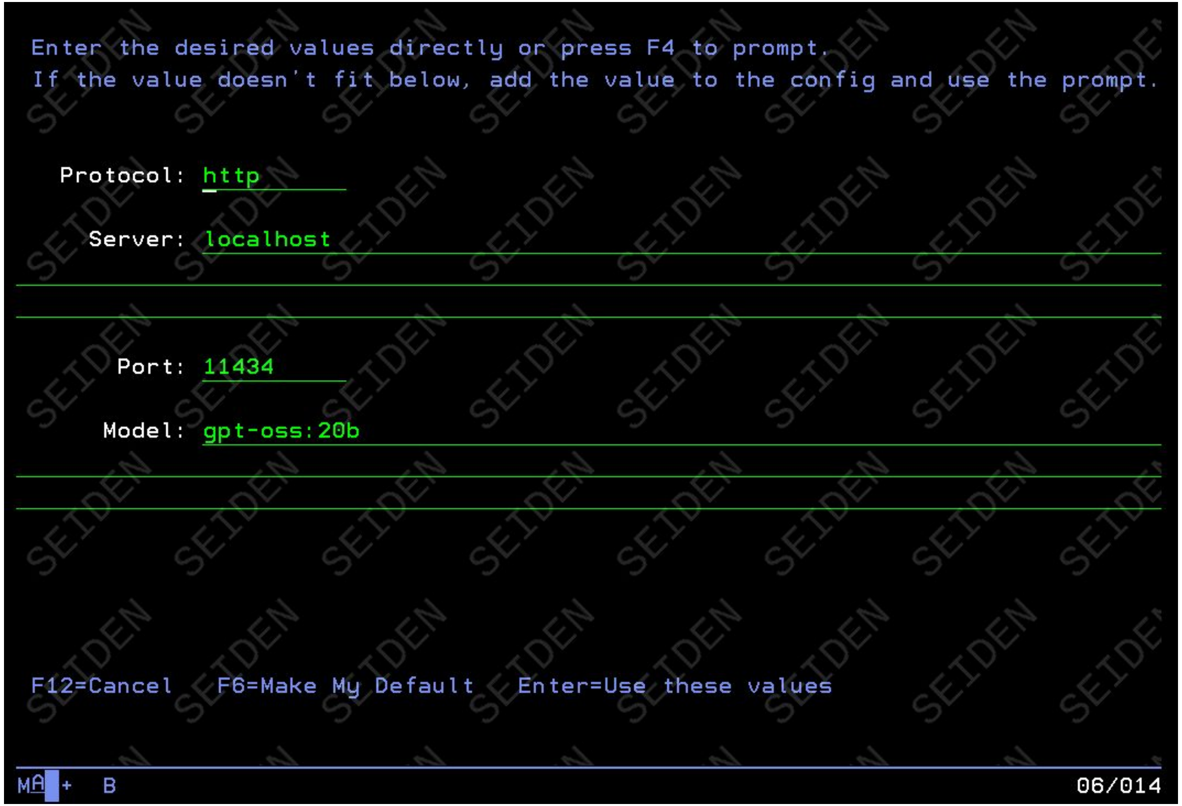
dcl-proc qNewLLM;
// Configure Ollama server settings
 exec sql call dbsdk_v1.ollama_setserverforjob(hostname => 'localhost');
 exec sql call dbsdk_v1.ollama_setportforjob(port => '11434');
 exec sql call dbsdk_v1.ollama_setprotocolforjob(protocol => 'http');
 exec sql call dbsdk_v1.ollama_setmodelforjob(model => 'llama3.2:3b'); // Using llama3.2:3b as per user's available model

 // Generate response from LLM
 exec sql select dbsdk_v1.ollama_generate(prompt => :PromptText) into :MyAnswer from SYSIBM.SYSDUMMY1;
 if SQLSTATE = '00000';
   DspStr = MyAnswer_DATA;
   dsply DspStr;
   snd-msg *INFO DspStr %target(*CALLER);
 else;
   DspStr = 'SQL Error: ' + SQLSTATE;
   dsply DspStr;
   snd-msg *INFO DspStr %target(*CALLER);
 endif;

 *inlr=*on;
end-Proc;
```

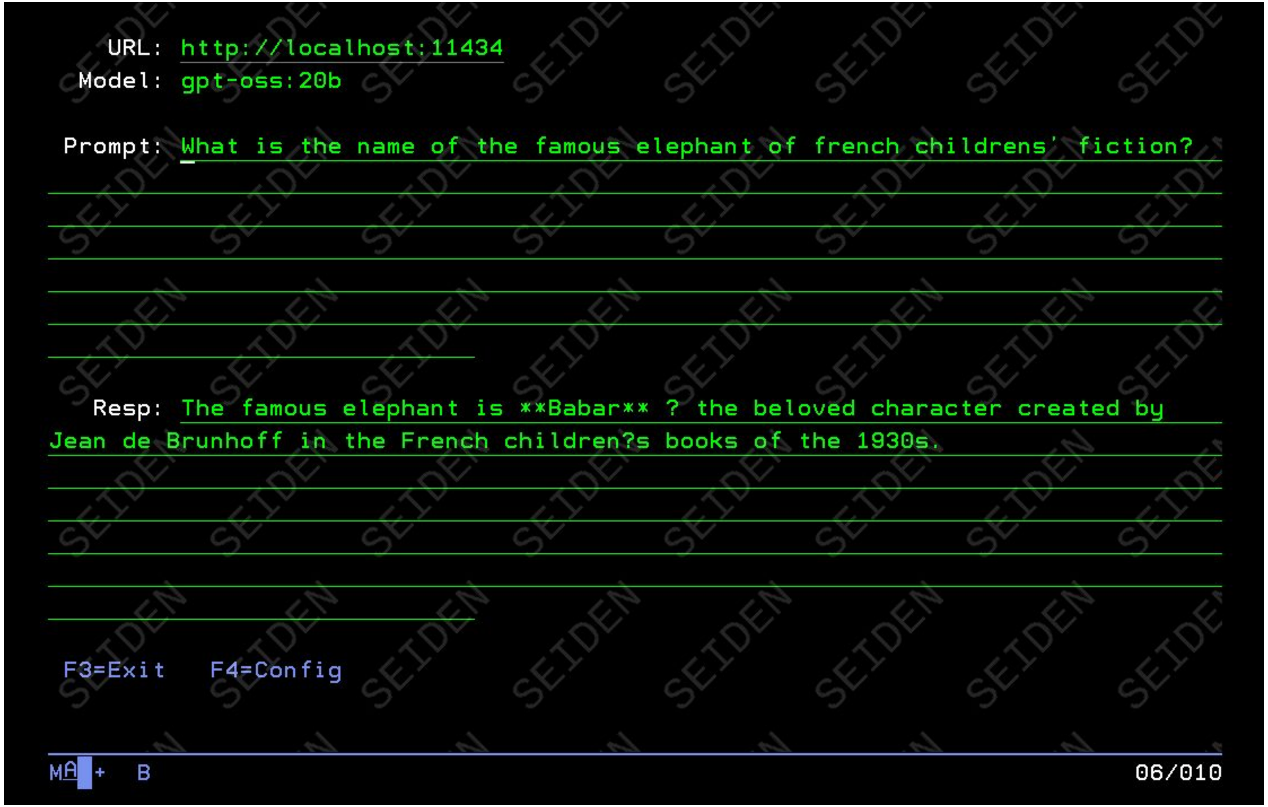
AI via DSPF

- **OLLAMARPG**
- GEMQRY



AI via DSPF

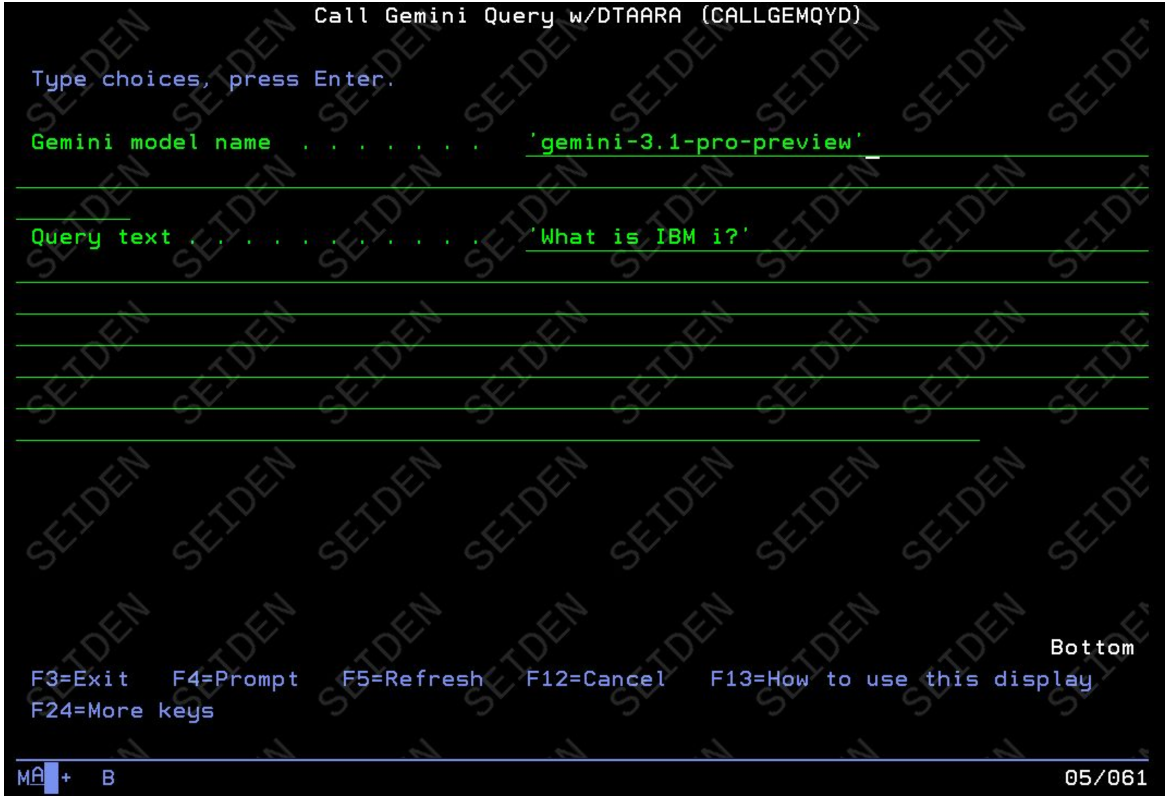
- **OLLAMARPG**
- GEMQRY



AI via DSPF

- OLLAMARPG
- **GEMQRY**

```
// Call the GEMINI_QUERY function
EXEC SQL
 SET :hResult = GEMINI_QUERY(
   MODEL => TRIM(:hModel),
   KEY => TRIM(:hKey),
   CONTENT => TRIM(:hContent)
 );
```



AI via DSPF

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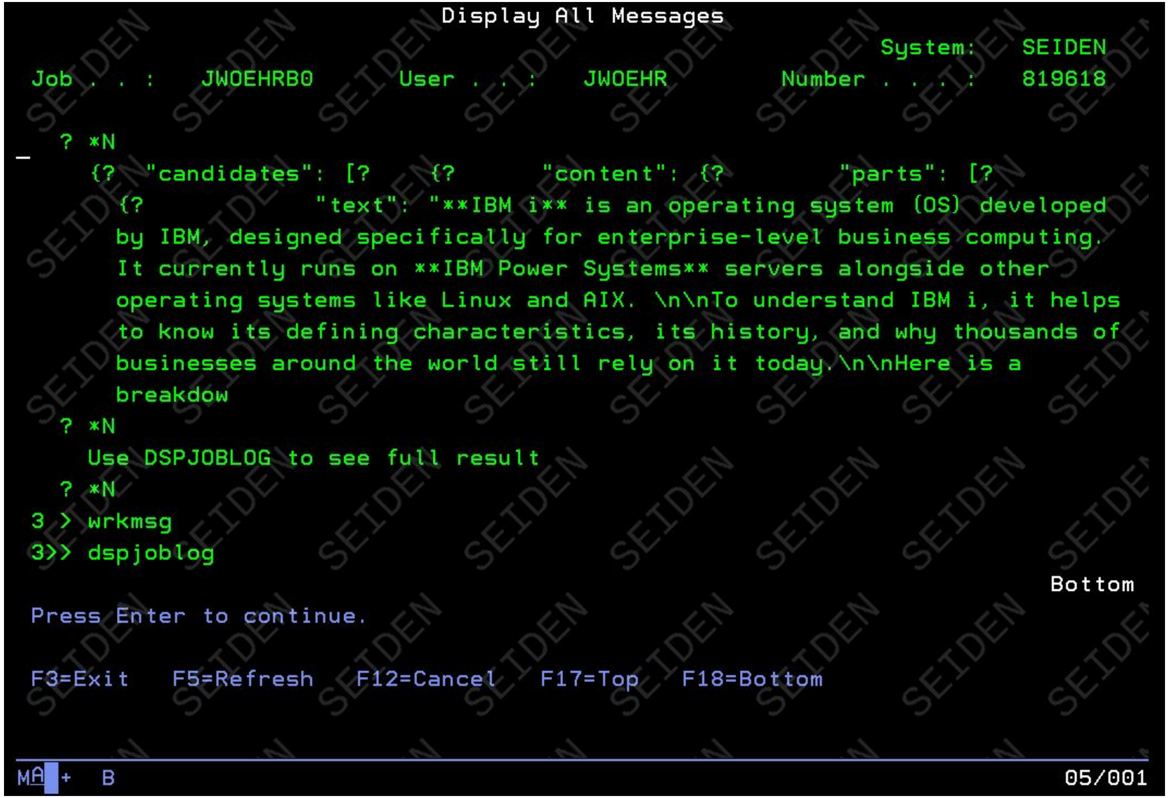
```



The code (and the PDFs for the presentation)

Thank you

Any sufficiently advanced technology is indistinguishable from magic. - A. C. Clarke

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